

Milestone 4 Technical Summary

Architecture and End-to-End Behavior

The system now implements two real workflows. Search mode retrieves and ranks opportunities without pretending the search sentence is a posting. Posting mode classifies a job description, retrieves comparable evidence, and applies deterministic missing-data and authorization gates.

Layer	Method	Business control
Mode	search vs. posting	Prevents task confusion
Retrieval	80k index + hard constraints + dedupe	Inspectable candidates
Decision	word+character TF-IDF LinearSVC	0.984 macro F1
Policy	calibrated margin + deterministic gates	Review / Hold
Response	evidence-bound template	No unsupported facts

Evaluation vs. Baseline

System	Rows	Accuracy	Macro F1
M2 same-subset baseline	4,000	-	0.791
M3 true PEFT LoRA	4,000	0.875	0.873
M4 trained decision layer	4,000	0.984	0.984

The same-subset macro-F1 gain is +0.192. The model was trained on 77,135 rows after exact-text deduplication, not only the 2,000 rows used by M3 LoRA.

Robustness Audit

The original random split contains 1,328 exact validation texts that also appear in training. Macro F1 remains 0.979 on novel texts, 0.980 for unseen companies, and 0.976 on the newest 20,000 postings. A disjoint 10,000-row policy audit retains 91.6% coverage at 99.47% selective accuracy.

Grounded Workflow Evidence

Case	System result	Operational response
entry level analytics search	N/A (search)	Advisor review before forwarding
senior engineering posting	low_fit	Investigate - advisor review required
authorization restriction posting	high_fit	Hold - explicit authorization restriction requires advisor verification
thin metadata posting	unclear	Investigate - advisor review required

Risk and Grounding Analysis

Risk 1 is authorization misinformation. Role fit and authorization are separate: explicit negative language forces a Hold, and missing evidence never becomes eligibility. Risk 2 is weak-label overconfidence. The high score measures agreement with transparent rules, so the system preserves human review and reports novel-text and unseen-company slices rather than relying on a random split alone.

Completed Validation Package

The repository includes a full 785,741-row source scan, 230 authorization candidates, a 430-row benchmark, and a 1,000-row advisor queue. It also includes a calibrated review policy, a six-query search audit with 0 violations across 36 results, and a batch CSV workflow. Human fields remain blank for genuine advisor review.

Deployment Boundary

Use the prototype for first-pass prioritization. Search results, low-margin classifications, missing fields, and any authorization-sensitive case remain advisor-confirmed. No automated forwarding, rejection, or immigration decision is permitted.